

DELCODE-PET | Qualitative ratings of Florbetaben amyloid PET

Authors: Marcel Daamen, Henning Boecker

E-Mail: PET-group@dzne.de

Last update: 17-Jun-2022

Purpose:

This document provides background information about the visual ratings of the Neuraceq™ (18F-Florbetaben [FBB]: Life Radiopharma, Berlin) amyloid PET scans which were performed in the DELCODE-PET study.

These qualitative image analyses were subject to a research agreement with Life Molecular Imaging. Moreover, some Florbetaben PETs were conducted in cooperation with the AMYPAD Consortium. Therefore, article or poster acknowledgments which use this information should contain the information: *“The qualitative ratings of the Florbetaben PET exams reported in this study were subject of a research agreement with Life Molecular Imaging. Some Florbetaben PET examinations were performed with support of the AMYPAD consortium”*.

Analyzed FBB data:

In general, FBB PET examinations were **only collected from Subjective Cognitive Decline (SCD) participants from the DELCODE study** who agreed to these additional examinations.

No complementary PET data are available from healthy controls, MCI and AD dementia participants!

FBB PET examinations were **performed up to two times**: A baseline scan (“FBB_P0”) and a follow-up scan (“FBB_P24”).

Note: Due to the delayed start of DELCODE-PET, the baseline scan was not always performed at the baseline visit of the DELCODE main study, but during follow-up visits of the DELCODE main study. Accordingly, the “P0” and “P24” labels for the PET data are not equivalent with the “M0, M12, M24, ...” labels found in the MRI data. Moreover, while follow-up PET scans were originally scheduled for a 24-month delay (hence: “P24”), this was later extended to a time window of up to 36 months.

Data acquisition:

PET data acquisition followed established standard protocols:

	FBB
Applied tracer activity	260-300 Megabequerel (MBq)
Start of emission scan	90 min post-injection
Duration of emission	20 min
Acquisition mode	3D Listmode, <i>post hoc</i> reconstruction in 4*5 min time frames
Reconstruction algorithm	<i>Iterative:</i> Based on established brain protocols at PET study sites • <i>Siemens + GE:</i> Ordered Subset Expectation Maximization (OSEM) • <i>Philips:</i> Line-of-response row-action maximum likelihood (LOR-RAMLA)
Image corrections	• Decay • Randoms • Dead time • Scatter • Attenuation • Normalization
Attenuation correction	Based on vendor-specific attenuation coefficient (μ) maps • <i>Siemens/ GE/ Philips PET/CT systems:</i> Derived from low-dose CT • <i>Siemens PET/MR systems:</i> Derived from DIXON-VIBE MR sequence

Data analysis:

To prepare for qualitative ratings, the 4 x 5 min time frames of the original attenuation-corrected iterative reconstructions were motion corrected and averaged using PMOD (PMOD Technologies LLC, Zurich, Switzerland).

Qualitative ratings of amyloid positivity were based on **consensus judgments of two readers with clinical experience in nuclear medicine**.

Based on guidelines of the *Neuraceq™ Educational Programme for Healthcare Professionals* (provided by Life Molecular Imaging), qualitative rating procedures included:

1. **A priori evaluation of technical interpretability (yes/ limited/ no):** General check for visible motion artifacts, suboptimal attenuation correction, atrophy, etc. Scans rated “no” will not be provided.
2. **Qualitative ratings were primarily based on the PET emission data.** If necessary, complementary low-dose CT or MR image data were consulted (e.g. to control potential influences of atrophic changes)
3. **Two-level qualitative rating of regional tracer binding:**
 - As described earlier (e.g. [Barthel et al., Lancet Neurology 2011](#))
 - *First Level:* Scoring of Regional Cortical Tracer Uptake (RCTU) for 4 predefined brain regions (Figure 1):

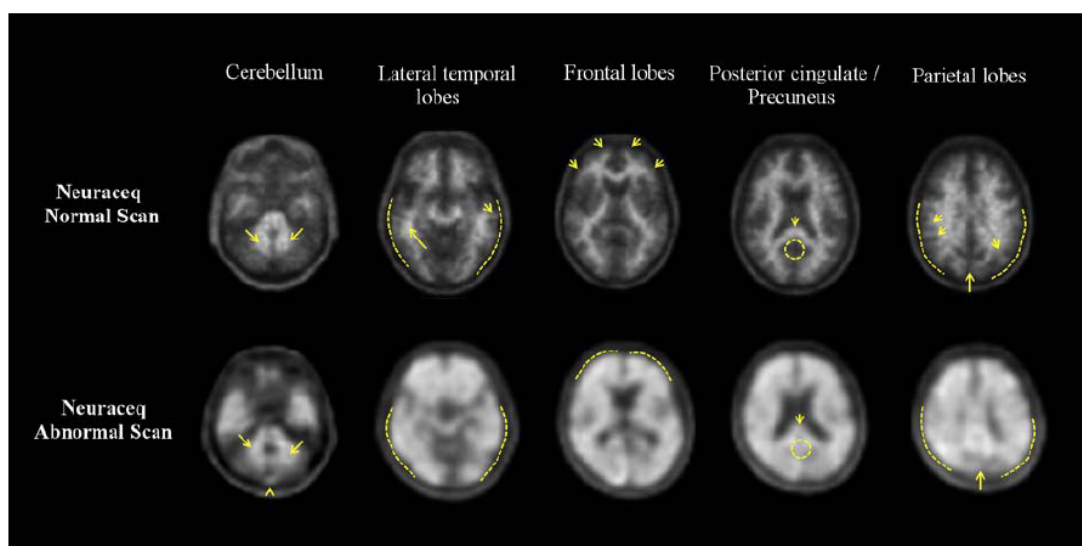


Figure 1: Brain regions for RCTU assessment ([European Medicines Agency](#))

RCTU	Meaning	Explanation
1	No Uptake	No cortical uptake in the majority of slices representing the brain region
2	Moderate Uptake	Smaller areas with cortical uptake, but in the majority of image slices representing the brain region
3	Pronounced Uptake	Large confluent area of cortical uptake, but in the majority of image slices representing the brain region

— *Second Level: Assignment of Brain Amyloid Plaque Load (BAPL)*

- Global evaluation of amyloid plaque burden
- Derived from previous RCTU scorings
- Three-level classification of beta-amyloid deposition
- Can be simplified to a positive/ negative categorization

Judgment:	BAPL-Score		Conditions
Negative	1	No beta-amyloid deposition	RCTU = 1 for all 4 regions
Positive	2	Moderate beta-amyloid deposition	RCTU = 2 for at least one of 4 regions, but no region with RCTU=3
	3	Pronounced beta-amyloid deposition	RCTU = 3 for at least one of 4 brain regions

4. **Rating confidence:** Raters provided a judgment of their rating confidence

Variables:

Based on the abovementioned procedures, the following main outcome variables are provided...

FBB_QUAL_P0 FBB_QUAL_P24	<i>FBB Visual data interpretation</i> Binary classification of FBB PET: Amyloid-positive/ -negative
FBB_BAPL_P0 FBB_BAPL_P24	<i>FBB BAPL score</i> Three-level classification of <u>b</u> rain <u>a</u> myloid <u>p</u> laque <u>l</u> oad: 1- negative 2- moderate 3 - pronounced
FBB_SURE_P0 FBB_SURE_P24	<i>Rating confidence:</i> 1= low 2=moderate 3=strong

Moreover, FBB qualitative ratings come along with additional background variables:

FBB_QUAL_COMM_P0 FBB_QUAL_COMM_P24	<i>Quality comments</i>
FBB_DATE_P0 FBB_DATE_P24	<i>Date of PET examination</i>
FBB_REF_P0 FBB_REF_P24	<i>Clinical reference visit</i> Especially for FBB, PET exam dates may fall in-between two consecutive DELCODE main study visits. Here, the study visit <i>closer in time</i> is suggested (which can be the next one).
AGE_FBB_P0 AGE_FBB_P24	<i>Age at time of FBB PET examination</i>
FBB_DOSE_P0 FBB_DOSE_P24	<i>Decay-corrected injected tracer activity [in Megabequerel, MBq]</i> For some participants, the injected FBB tracer dose was lower than defined in the DELCODE protocol (260-300 MBq), but still consistent with manufacturer's suggestions (300±60 MBq) in most cases. Tracer over- and underdosings were not generally excluded if image quality was sufficient for visual interpretation, but this variable can be used for selecting participants more stringently.

FBB_DELAY_P0 FBB_DELAY_P24	<i>Time between tracer injection and start of PET emission [in min]</i> Based on available DICOM and checklist information. For some participants, emission scan did not start regularly, i.e. 90 min post-injection, usually due to technical problems. These cases were not generally excluded, if image quality was sufficient for visual interpretation, but this variable can be used for selecting participants more stringently.
FBB_PETCAM_P0 FBB_PETCAM_P24	<i>Vendor and model of PET scanner</i> • GE Discovery 690: PET/CT • Philips Gemini TF 16 TOF: PET/CT (@two sites) • Siemens Biograph Duo: PET/CT • Siemens Biograph mCT 128 Flow Edge: PET/CT • Siemens Biograph mMR: PET/MRI (@two sites)
FBB_RECON_P0 FBB_RECON_P24	<i>Original PET reconstruction algorithm</i> In case that detailed technical information is needed for study methods, information is provided here.
FBB_GEO_P0 FBB_GEO_P24	<i>Original PET image geometry</i> In case that technical information is needed for study methods, information is provided here.
FBB_DEVICE_P0 FDG_DEVICE_P0 FBB_DEVICE_P24 FDG_DEVICE_P24	<i>PET scanner ID</i> In case that technical information is needed for study methods, information is provided here. For optional use of scanner identity as a statistical covariate. Note that some clinical sites used the same PET scanner. Identical with <i>Device Serial Number</i> (0018,1000) in DICOM header. Because the original vendor information was already deleted by some PET sites before upload, and for pseudonymization, each single scanner received a new, arbitrary ID string ("00001"- "00007").